

ASSESSMENT TOOL 1: Analytical Problem Solving
COURSE NAME: COMPUTER VISION
COURSE CODE: ITA0515
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1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

Answer: Grey level transformation changes the intensity values of image pixels to improve brightness, contrast, or visibility of details.

(b) Apply a negative transformation $s = 255 - r$ to the pixel values: $r = [50, 100, 150, 200]$. Compute the transformed values.

Answer:

$$s = 255 - r$$

For 50: $255 - 50 = 205$

For 100: $255 - 100 = 155$

For 150: $255 - 150 = 105$

For 200: $255 - 200 = 55$

Transformed values = $[205, 155, 105, 55]$

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

Answer: Log transformation expands low-intensity pixel values and compresses high-intensity values. Therefore, dark details become more visible.

(b) Given $s = c \log(1 + r)$, where $c = 10$, compute s for $r = [0, 10, 50, 100]$.

Answer:

Using log base 10:

For $r = 0$: $s = 10 \log_{10}(1 + 0) = 0.00$

For $r = 10$: $s = 10 \log_{10}(1 + 10) = 10.41$

For $r = 50$: $s = 10 \log_{10}(1 + 50) = 17.08$

For $r = 100$: $s = 10 \log_{10}(1 + 100) = 20.04$

Transformed values $\approx [0.00, 10.41, 17.07, 20.04]$

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

Answer: Histogram equalization redistributes pixel intensities so that the image uses a wider range of intensity levels. This improves contrast, especially when the image is too dark or too bright.

(b) Given a 3-bit image with histogram:

Intensity	Frequency
0	2
1	2
2	4
3	8

Answer:

Total pixels = $2 + 2 + 4 + 8 = 16$

Cumulative frequencies: $[2, 4, 8, 16]$

For a 3-bit image, $L = 8$. Using $s = \text{floor}((L - 1) \times \text{CDF} / 16)$:

Intensity 0: $\text{floor}(7 \times 2/16) = 0$

Intensity 1: $\text{floor}(7 \times 4/16) = 1$

Intensity 2: $\text{floor}(7 \times 8/16) = 3$

Intensity 3: $\text{floor}(7 \times 16/16) = 7$

Equalized intensity values = [0, 1, 3, 7]

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

Answer: Smoothing filters reduce noise and small variations in an image. They produce a smoother image by averaging neighboring pixel values.

(b) Apply a 3×3 averaging filter to the region:

10	20	30
20	30	40
30	40	50

Answer:

$$\begin{aligned}\text{New center pixel} &= (10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50) / 9 \\ &= 270 / 9 = 30\end{aligned}$$

New center pixel value = 30

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

Answer: Sharpening emphasizes edges and fine details by increasing the difference between a pixel and its neighboring pixels.

(b) Apply the Laplacian mask:

0	-1	0
-1	4	-1
0	-1	0

to the matrix:

10	10	10
10	20	10
10	10	10

Answer:

$$\text{Output at center} = (0 \times 10) + (-1 \times 10) + (0 \times 10) + (-1 \times 10) + (4 \times 20) + (-1 \times 10) + (0 \times 10) + (-1 \times 10) + (0 \times 10)$$

$$= -10 - 10 + 80 - 10 - 10 = 40$$

$$\text{Output at the center} = 40$$